AI Use Appendix

Working Status

This appendix now reflects the completed verification and final canonical OpenAI-powered run. It documents representative development prompts, selected output excerpts, editorial judgment, and the verification steps that were accepted, revised, rejected, or independently checked.

1. Representative Development Prompts

Keep this appendix selective. Include representative prompts that materially changed the implementation.

Prompt A: Base architecture

Goal:

- Build a LangGraph workflow with a pure market-data node, independent strategy branches, and evaluator routing.

Representative summary:

> *Design a LangGraph workflow where market data fans out to independent strategy agents, those branches rejoin before evaluation, and the system handles agreement and disagreement differently.*

What I accepted:

- the graph-first framing of the assignment

What I revised:

- I used schema-constrained outputs instead of free-form text parsing

Prompt B: Indicator selection

Goal:

- Turn the assignment's behavioral strategies into concrete indicators.

Representative summary:

> *Compare a Momentum Trader against a Value Contrarian using moving averages, recent returns, volume confirmation, RSI, drawdown, 52-week range distance, and trailing P/E.*

What I accepted:

- moving-average structure as the center of the momentum lens

What I revised:

- `trailing_pe` became an explicit valuation anchor in the contrarian design

Prompt C: Bonus extensions

Goal:

- Add all three optional extensions without turning the code into a rewrite.

Representative summary:

> *Add a third strategy agent, disagreement-only debate mode, and a historical backtest scorecard while keeping the online workflow clean and the historical logic separate.*

What I accepted:

- keeping backtest logic outside the LangGraph online path

What I revised:

- I made debate a disagreement-only second round and intentionally omitted live valuation data from historical snapshots to avoid lookahead bias

2. Representative Output Excerpts

Excerpt A: Live market-only stock-selection evidence

Observed on 2026-04-10:

- `COST`: `pct_change_30d = 1.19`, `short_vs_long_ma_pct = 0.27`
- `HIMS`: `pct_change_30d = 24.55`, `short_vs_long_ma_pct = 5.68`
- `SMCI`: `pct_change_30d = -21.75`, `short_vs_long_ma_pct = -14.31`

- `JNJ`: `pct_change_30d = -2.06`, `short_vs_long_ma_pct = -0.32`
- `TSLA`: `pct_change_30d = -14.59`, `short_vs_long_ma_pct = -5.95`

Why it mattered:

- the final stock basket was chosen from live data rather than intuition alone

Excerpt B: Strategy excerpt

> *"the setup looks more like a steady hold than an actionable breakout."*

Why it mattered:

- This was the clearest example of productive debate. The Momentum Trader did not simply defend its first answer; it reconsidered the evidence and downgraded COST from `BUY` to `HOLD`.

Excerpt C: Evaluator excerpt

> *"not oversold at RSI 45.1, not trend-confirmed, and still structurally damaged despite the rebound."*

Why it mattered:

- This captured the central analytical payoff of the assignment in an agreement case: the evaluator explained why the rebound in HIMS was still too weak to convince any of the strategies.

3. Editorial Judgment

Accepted

- the LangGraph branch-and-routing architecture
- the use of OpenAI Structured Outputs plus Pydantic
- the separation between live workflow logic and historical backtest logic

Revised

- I revised the project to make the core contrast explicitly `P/E` versus moving-average structure

- I added a third strategy that brings a risk-first lens rather than a minor prompt variation
- I revised historical snapshots to omit live valuation fields and avoid lookahead bias
- I tested `NVDA` as a fifth live stock, rejected it because it produced another low-insight consensus case, and replaced it with `TSLA` for the final canonical basket

Rejected

- any design where one strategy sees another before the first-round evaluation
- free-form JSON parsing without typed validation
- historical backtest logic that reuses current valuation data as if it were historical

Editorial corrections applied to output files

Two LLM generation artifacts were identified during post-run review and corrected editorially before submission.

Artifact 1 — `outputs/HIMS.json`, `strategy_b.justification`:

The Value Contrarian justification contained a mid-sentence self-correction: `"The stock is also above its 20-day moving average of 21.29? Actually it is below that at 19.43"`. The final conclusion was factually correct (current price 19.43 is below the 20-day MA of 21.29), but the phrasing shows the model reconsidering mid-sentence rather than expressing a clear conclusion. The sentence was rewritten as: `"The stock is below its 20-day moving average of 21.29 at current price 19.43, and also below the 50-day average of 20.14"`. No data or conclusion was changed.

Artifact 2 — `outputs/backtest.json`, COST checkpoint Volatility Averse justification:

Same class of artifact: `"but it still sits below the 50-day moving average at 894.59? Actually current price is above the 50-day but below the 200-day at 950.25"`. The fact (price 924.88 is above the 50-day MA at 894.59 and below the 200-day at 950.25) is correct. The sentence was rewritten as: `"but current price of 924.88 is above the 50-day moving average at 894.59 while still below the 200-day at 950.25"`.

Prompt update to prevent recurrence:

Each of the three strategy prompts (`strategy_a.txt`, `strategy_b.txt`, `strategy_c.txt`) was updated to include: `"Write each sentence in your justification as a clear, direct statement; do not revise or self-correct mid-sentence."` This does not change strategy philosophy or decision logic; it addresses the output writing style only.

Verified Independently

- the market-data pipeline works with live Yahoo Finance data
- the upgraded workflow compiles and renders as a Mermaid graph
- the installed OpenAI SDK exposes `client.responses.parse(...)`
- the three-strategy disagreement and debate control flow works in a no-key smoke test
- the backtest scorecard logic works in a no-key synthetic smoke test

4. Verification Log

Already completed:

- `python3 -m venv .venv`
- `./.venv/bin/pip install -r requirements.txt`
- `./.venv/bin/python -m compileall src`
- `./.venv/bin/python -m src.main --market-only COST`
- direct live checks of candidate tickers through `fetch_market_data(...)`
- local inspection of `client.responses.parse(...)`
- deterministic smoke test for the extended graph and backtest logic

Final execution commands run:

- `./.venv/bin/python -m src.main --verify-setup`
- `./.venv/bin/python -m src.main`
- exploratory but non-canonical candidate scan was run separately in `/tmp` to test whether alternative live tickers would generate richer disagreement
- the final canonical batch was rerun after replacing `NVDA` with `TSLA`

5. Final Submission Checklist

- [x] one JSON file exists for each analyzed stock
- [x] `outputs/summary.json` exists
- [x] `outputs/backtest.json` exists
- [x] the report includes at least two direct excerpts from generated outputs
- [x] one failure or surprise case is documented honestly

- [x] prompts include `strategy_a.txt`, `strategy_b.txt`, `strategy_c.txt`, `evaluator.txt`, and `debate.txt`
- [x] README explains setup and the bonus-extension workflow
- [x] this appendix explains what was accepted, revised, rejected, and verified independently